VetLink PWA — Clinical & Feature Update Report

Project: A Deep Learning Animal Decision Support System Using MobileNetV3 and Grad-CAM for VetLink Clinic
Date: October 22, 2026 • **System Completion:** 95% • **Branch:** feature/ai-decision-support-v2

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1. Executive Summary

This milestone delivers four critical real-world clinical capabilities addressing veterinarian workflows, quality control, and temporal tracking:

- **Reconciliation of Upstream Merge Conflicts:** Cleanly integrated upstream sync endpoints with native TFLite AI endpoints in backend/main.py and App.css.
- **Interactive & Editable CDSS Prescription:** Replaced static protocol views with editable dosage and clinical notes boxes saved to IndexedDB.
- **Printable Rx Slip & PDF Export:** Browser-native window.print() and @media print styling generating formal veterinary prescription sheets.
- **Client-Side Image Blur Detection:** Real-time Laplacian edge gradient variance algorithm checks focus before AI inference.
- **Longitudinal Follow-Up / Recheck Workflow:** Side-by-side Before & After comparison with human-in-the-loop 4-stage healing progress evaluation.

2. Detailed Clinical Enhancements

Feature 1: Interactive & Editable Prescription Plan (Treatment.jsx)

The AI acts as Decision Support by auto-drafting recommendations based on diagnosis. The veterinarian has full clinical autonomy to adjust mg/kg dosage and custom pet-owner precautions. Clicking *Save & Finalize* updates the consultation record via updateConsultation() in IndexedDB and queues it for sync.

Feature 2: Official Printable Prescription Slip & PDF Export (App.css)

Clicking *Print Rx / Save PDF* automatically triggers browser printing. The dedicated print stylesheet hides digital navigation and outputs a formal Veterinary Rx slip with clinic letterhead, consultation date, patient info, validated diagnosis, ■ prescription, and clinician signature block.

Feature 3: Client-Side Image Sharpness / Blur Detection (vetlink-ai.js)

Calculates edge gradient variance on a 128x128 canvas. If sharpness score is below 12, a yellow Quality Alert appears beneath the photo preview advising the user to hold the camera steady and offering an instant *Retake Photo* button.

Feature 4: Longitudinal Recheck & Healing Stage Evaluation (Patients.jsx)

Addresses the time-blindness of computer vision models. Every patient row has a ■ **RECHECK** button opening a side-by-side comparison of the Day 0 Baseline photo vs. Today's photo. The veterinarian grades progress: ■ Resolved / Healed, ■ Responding / Improving, ■ Refractory / Unchanged, or ■ New Complication.

3. Key Academic Justifications for Capstone Panel

- **Why Datasets Have No Dates:** Image classifiers (MobileNetV3) operate on spatial morphology, not temporal dates. Time is an Electronic Health Record (EHR) variable managed by VetLink's database, not the CNN.
- **The 60% Confidence Decision Boundary (FR7):** Low confidence (<60%) indicates blurry images, occlusions (hands), or atypical overlapping lesions, safely triggering secondary laboratory diagnostics (Skin Scraping, Woods Lamp, Cytology).
- **Multi-Disease Comorbidity Storage:** Secondary differentials (≥25%) are preserved in screeningResults.aiTop3 (JSON) and recorded in the consultations.diagnosis text field for complete clinical auditability.

4. Verification Status

Test / Component	Command / Target	Result
Frontend Build	npm run build	PASS (444ms, 0 errors)
Backend Python	python -m py_compile backend/main.py	PASS (0 syntax errors)
Git Status	git status	Clean (ahead by 2 commits)
Completion Rate	VetLink System Evolution	95% Complete